Blog Authorship Corpus Analysis using NumPy and Pandas

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Aim

Toanalysethe Blog Authorship Corpus dataset using NumPy and Pandas libraries and solve real-world problem statements using data analysis techniques.

Objective

- To understandthe structure of real-life datasets.
- To perform data cleaning and preprocessing.
- To apply NumPy and Pandas operations.
- To perform statistical and analytical operations.
- To visualize insights from the dataset.

Dataset Description

TheBlogAuthorshipCorpus dataset containsinformation related to blog authors such as age, gender, topic, zodiac sign, and blog content. The dataset is useful for analysing blogging patterns, demographic trends, and topic preferences.

The dataset includes the following attributes:

- Gender
- Age
- Topic
- Zodiac Sign
- Blog Text
- Date

Libraries Used

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Matplotlib	Data visualization

Importing Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Loading Dataset

```
df = pd.read_csv("dataset/blog.csv")
```

Understanding the Dataset

```
print(df.head())
```

```
print(df.shape)
```

```
print(df.columns)
```

```
print(df.info())
```

```
print(df.describe(include='all'))
```

Interpretation

The dataset contains multiple records of blog authors with categorical and numerical data. Understanding the dataset structure helps in performing accurate analysis.

Data Cleaning

Checking Null Values

```
print(df.isnull().sum())
```

Removing Null Values

```
df = df.dropna()
```

Removing Duplicate Records

```
df = df.drop_duplicates()
```

Interpretation
Data cleaning improves data quality and ensures accurate analysis results.

Problem Statements and Solutions

Problem Statement 1

Findthetotalnumberofblog posts.

```
print(len(df))
```

Interpretation
This operation gives the total number of records available in the dataset.

Problem Statement 2

Findthetotalnumberofmaleand female bloggers.

```
print(df['gender'].value_counts())
```

Interpretation
This helps in understanding the gender distribution of bloggers.

Problem Statement 3

Find the average age of bloggers.

```
print(df['age'].mean())
```

Interpretation
The mean age represents the average age of bloggers present in the dataset.

Problem Statement 4

Findmaximumandminimumage of bloggers.

```
print(df['age'].max())
print(df['age'].min())
```

Interpretation
This operation identifies the oldest and youngest bloggers.

Problem Statement 5

Find the most common zodiac sign.

```
print(df['sign'].mode())
```

Interpretation

The mode function helps identify the most frequent zodiac sign.

Problem Statement 6

Count blogs written for each topic.

```
print(df['topic'].value_counts())
```

Interpretation

This shows which blogging topics are most popular.

Problem Statement 7

Findtopichavinghighestnumber of blogs.

```
print(df['topic'].value_counts().idxmax())
```

Interpretation

The result identifies the most active blogging category.

Problem Statement 8

Findaverageagetopic-wiseusing groupby().

```
print(df.groupby('topic')['age'].mean())
```

Interpretation

This analysis helps compare age trends across different topics.

Problem Statement 9

Find gender distribution topic-wise.

```
print(df.groupby(['topic', 'gender']).size())
```

Interpretation
This helps understand which gender contributes more in specific topics.

Problem Statement 10

Find total bloggers in each age group.

```
print(df.groupby('age').size())
```

Interpretation
This operation shows the distribution of bloggers according to age.

Problem Statement 11

Find top 10 most active topics.

```
print(df['topic'].value_counts().head(10))
```

Interpretation
This identifies the most discussed blogging topics.

Problem Statement 12

Findpercentageofmaleandfemale bloggers.

```
print((df['gender'].value_counts()/len(df))*100)
```

Interpretation
This operation converts gender distribution into percentage form.

Problem Statement 13

Find standard deviation of age.

```
print(np.std(df['age']))
```

Interpretation
Standard deviation measures the spread of age values.

Problem Statement 14

Find variance in age.

```
print(np.var(df['age']))
```

Interpretation
Variance indicates how much the ages differ from the mean age.

Problem Statement 15

Findbloggersolderthanaverage age.

```
print(df[df['age'] > df['age'].mean()])
```

Interpretation
This filters bloggers whose age is greater than the average age.

Problem Statement 16

Sort bloggers according to age.

```
print(df.sort_values(by='age'))
```

Interpretation
Sorting helps arrange bloggers in ascending order of age.

Problem Statement 17

Find unique blog topics.

```
print(df['topic'].unique())
```

Interpretation
This operation identifies all unique categories available in the dataset.

Problem Statement 18

Find total numberof unique topics.

```
print(df['topic'].nunique())
```

Interpretation
This gives the total count of distinct blog topics.

Problem Statement 19

Aggregate multiple statistics together.

```
print(df.groupby('gender')['age'].agg(['mean', 'max', 'min', 'count']))
```

Interpretation
Aggregation provides multiple statistical insights in a single operation.

Problem Statement 20

Apply NumPy conditional analysis.

```
print(np.where(df['age']> 30, 'Senior Blogger','Young Blogger'))
```

Interpretation
NumPy conditional analysis classifies bloggers based on age.

Data Visualizations

Visualization 1–GenderDistribution

```
df['gender'].value_counts().plot(kind='bar')
plt.title('Gender Distribution')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```

Interpretation
The graph visually represents the number of male and female bloggers.

Visualization 2 – Top Topics

```
df['topic'].value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Topics')
plt.xlabel('Topics')
plt.ylabel('Count')
plt.show()
```

Interpretation
This graph highlights the most popular blogging topics.

Visualization 3 – Age Distribution

```
df['age'].plot(kind='hist')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.show()
```

Interpretation
The histogram shows how blogger ages are distributed.

Conclusion

The Blog Authorship Corpus dataset was successfully analysed using NumPy and Pandas libraries. Various operations such as filtering, grouping, aggregation, sorting, statistical analysis, and visualization were performed. The analysis provided useful insights into blogger demographics, age patterns, gender distribution, and topic popularity.

This assignment helped in understanding practical applications of NumPy and Pandas for real-world data analysis.
